

The Mean, the Median, and the Mode

3 Different Ways to Find the Average of Something

by Deb Russell Updated February 09, 2017

Before you can begin to understand statistics, there are four terms you will need to fully understand. The first term 'average' is something we have been familiar with from a very early age when we start analyzing our marks on report cards. We add together all of our test results and then divide it by the sum of the total number of marks there are. We often call it the average. However, statistically it's the Mean!

THE MEAN

Example:

Four tests results: 15, 18, 22, 20

The sum is: 75

Divide 75 by 4: 18.75

The 'Mean' (Average) is 18.75

(Often rounded to 19)

THE MEDIAN

The Median is the 'middle value' in your list. When the totals of the list are odd, the median is the middle entry in the list after sorting the list into increasing order. When the totals of the list are even, the median is equal to the sum of the two middle (after sorting the list into increasing order) numbers divided by two. Thus, remember to line up your values, the middle number is the median! Be sure to remember the odd and even rule.

Examples:

Find the Median of: 9, 3, 44, 17, 15 (Odd amount of numbers)

Line up your numbers: 3, 9, 15, 17, 44 (smallest to largest)

The Median is: 15 (The number in the middle)

Find the Median of: 8, 3, 44, 17, 12, 6 (Even amount of numbers)

Line up your numbers: 3, 6, 8, 12, 17, 44

Add the 2 middle numbers and divide by 2: $8 + 12 = 20 \div 2 = 10$

The Median is 10.

THE MODE

The mode in a list of numbers refers to the list of numbers that occur most frequently. A trick to remember this one is to remember that mode starts with the same first two letters that most does. Most frequently - Mode. You'll never forget that one!

Examples:

Find the mode of:

9, 3, 3, 44, 17, 17, 44, 15, 15, 15, 27, 40, 8,

Put the numbers in order for ease:

3, 3, 8, 9, 15, 15, 15, 17, 17, 27, 40, 44, 44,

The Mode is 15 (15 occurs the most at 3 times)

*It is important to note that there can be more than one mode and if no number occurs more than once in the set, then there is no mode for that set of numbers.

THE RANGE

Occasionally in Statistics you'll be asked for the 'range' in a set of numbers. The range is simply the the smallest number subtracted from the largest number in your set. Thus, if your set is 9, 3, 44, 15, 6 - The range would be $44-3=41$. Your range is 41.

A natural progression once the 3 terms in statistics are understood is the concept of probability. Probability is the chance of an event happening and is usually expressed as a fraction. But that's another topic!

Edited by Anne Marie Helmenstine, Ph.D.

How to Find The Mean, Median, Mode and Range of a Series Of Numbers

by Todd Helmenstine
Updated March 24, 2017

When processing data, terms such as mean, median, mode, and range often come up. What do these terms mean and how do you calculate them? This worked example problem defines each term and demonstrates how to calculate the value of each term from a series of numbers.

MEAN, MEDIAN, MODE, AND RANGE DEFINITIONS

First, let's review what these terms mean:

Mean: The mean is the average of a set of numbers, determined by adding them together and dividing by how many numbers are present.

Median: The median is the middle value between the smallest and largest of a set of numbers.

Mode: The mode is the number that appears most often in a data set. It only has significance if numbers are repeated.

Range: The range is the difference between the largest and smallest numbers.

MEAN, MEDIAN, MODE, AND RANGE PROBLEM

Find the mean, median, mode, and range of the following set of values:

11, 19, 13, 16, 12, 12, 18, 14, 20

Solution:

It is often helpful to arrange the numbers from lowest to highest:

11, 12, 12, 13, 14, 16, 18, 19, 20

PART 1 - HOW TO FIND THE MEAN

The mean of a set of numbers is the average. The mean is calculated by finding the sum of all the values and dividing by the number of values.

$$11+12+12+13+14+16+18+19+20 = 135$$

There are 9 numbers in the series, so the mean is:

$$\text{Mean} = 135/9 = 15$$

A quick way to check if your answer is reasonable is to see if your answer is somewhere between the lowest and greatest number in the series.

In our case, 11 is the lowest and 20 is the greatest. 15 is between the two.

PART 2 - HOW TO FIND THE MEDIAN

The median of a series of numbers is the number that appears in the middle of the list when arranged from smallest to largest. For a list with an odd number of members, the way to find the middle number is to take the number of members and add one.

Then divide that value by two. In our case, there are 9 numbers in the series. $9+1 = 10$ and half of 10 is 5. The fifth number in the series is the median or 14.

If the number of members of the series was even, the average of the two middle numbers would be the median.

PART 3 - HOW TO FIND THE MODE

The mode is the number in the series that appears the most often. If there is no single number that appears more than any other number in the series, there is no value for the mode.

The number 12 appears twice in the series. The mode of this series is 12.

PART 4 - HOW TO FIND THE RANGE

The range is the difference between the largest number and the smallest number. The range is an indication of how spread out the values of the series are. The smaller the range, the closer the numbers in the series are to each other.

The range of our series is $20-11 = 9$.

Answer:

For the series:

11, 12, 12, 13, 14, 16, 18, 19, 20

The mean is 15. The median is 14. The mode is 12 and the range is 9.

To summarize:

Mean = average value

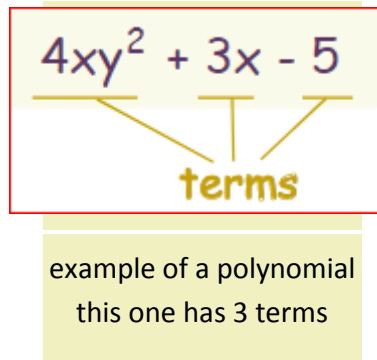
Median = middle value

Mode = most common value

Range = the "spread" of the series. This is the difference between largest and smallest numbers in the series.

Multiplying Polynomials

A polynomial looks like this:



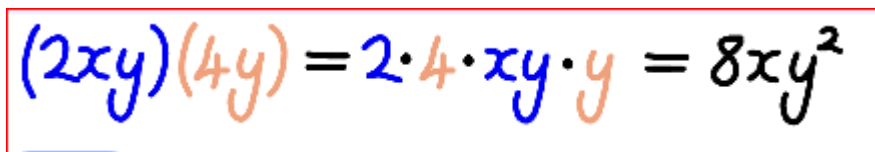
The diagram shows the polynomial $4xy^2 + 3x - 5$ enclosed in a red rectangular box. Three yellow horizontal lines are drawn under each term: $4xy^2$, $3x$, and -5 . Three yellow lines originate from these underlines and converge at a central point below the box, where the word "terms" is written in yellow. Below the box, a yellow rectangular box contains the text "example of a polynomial" and "this one has 3 terms".

To multiply two polynomials:

- multiply **each term** in one polynomial by **each term** in the other polynomial
- add those answers together, and simplify if needed

Let us look at the simplest cases first.

1 term $\times$ 1 term (monomial times monomial)



The equation $(2xy)(4y) = 2 \cdot 4 \cdot xy \cdot y = 8xy^2$ is shown in a red rectangular box. The numbers 2 and 4 are blue, the variables x and y are blue, and the second y is orange. The result 8xy^2 is black.

To multiply one term by another term, first multiply the **constants**, then multiply **each variable** together and combine the result, like this (press play):

(Note: I used "." to mean multiply. In Algebra we don't like to use "x" because it looks too much like the letter "x")

For more about multiplying terms, read [Multiply and Divide Variables with Exponents](#)

1 term $\times$ 2 terms (monomial times binomial)

Multiply the single term by each of the two terms, like this:

$$\underline{2x}(\underline{x} + \underline{3xy}) = \underline{2x} \cdot \underline{x} + \underline{2x} \cdot \underline{3xy} \\ = 2x^2 + 6x^2y \quad \checkmark$$

2 term $\times$ 1 terms (binomial times monomial)

Multiply each of the two terms by the single term, like this:

$$(\underline{x^2} - \underline{x})\underline{3y} = \underline{3x^2y} - \underline{3xy} \quad \checkmark$$

(I did that one a bit faster by multiplying in my head before writing it down)

2 terms $\times$ 2 terms (binomial times binomial)

Each of the **two terms** in the first binomial ...

... is **multiplied by** ...

... each of the **two terms** in the second binomial

That is 4 different multiplications ... Why?



Matching up Partners

Two friends (Alice and Betty) challenge two other friends (Charles and David) to individual tennis matches.

How many matches does that make?

- Alice needs to play Charles, and then Alice needs to play David.
- And Betty has to play Charles and then Betty has to play David.

They could play in any order, so long as **each of the first two friends** gets to play **each of the second two friends**.

It is the same when we are multiplying binomials!

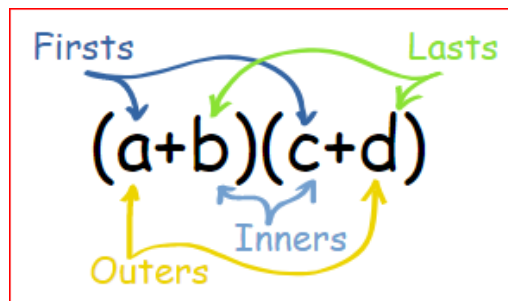
Instead of Alice and Betty, let's just use **a** and **b**, and Charles and David can be **c** and **d**:

$$(a+b)(c+d) = ac + ad + bc + bd$$

We can multiply them in any order so long as **each of the first two terms** gets multiplied by **each of the second two terms**.

But there is a handy way to help us remember to multiply each term called "**FOIL**".

It stands for "**F**irsts, **O**uters, **I**nners, **L**asts":



- Firsts: **ac**
- Outers: **ad**
- Inners: **bc**
- Lasts: **bd**

So you multiply the "Firsts" (the first terms of both polynomials), then the "Outers", etc.

Let us try this on a more complicated example:

$$(2x+3)(xz-a) = \overset{F}{2x^3z} - \overset{O}{2xa} + \overset{I}{3xz} - \overset{L}{3a}$$

2 terms $\times$ 3 terms (binomial times trinomial)

"FOIL" won't work here, because there are more terms now. But just remember:

$$(\underline{x} + \underline{a})(\underline{2x} + \underline{3y} - \underline{5}) = 2x^2 + 3xy - 5x + 2ax + 3ay - 5a$$

Multiply each term in the first polynomial by each term in the second polynomial

Like Terms

And always remember to add [Like Terms](#) :

Example: $(x + 2y)(3x - 4y + 5)$

$$\begin{aligned} & (x + 2y)(3x - 4y + 5) \\ &= 3x^2 - 4xy + 5x + 6xy - 8y^2 + 10y \\ &= 3x^2 + 2xy + 5x - 8y^2 + 10y \end{aligned}$$

Note: $-4xy$ and $6xy$ are added because they are Like Terms.

Also note: $6yx$ means the same thing as $6xy$

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Question 1

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Polynomials (Algebra 1, Easy)

[Help](#)

Multiply out $(3x + 2)(4x - 5)$

A $12x^2 - 7x - 10$

B $12x^2 - 23x - 10$

C $12x^2 + 7x - 10$

D $12x^2 - 10$

Well Done!

+0.50

Multiply each term in one polynomial by each term in the other polynomial:

$$\begin{aligned}\text{So } (3x + 2)(4x - 5) \\ &= 3x(4x - 5) + 2(4x - 5) \\ &= 12x^2 - 15x + 8x - 10 \\ &= 12x^2 - 7x - 10\end{aligned}$$

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Question 2

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Polynomials (Algebra 1, Easy)

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What is the product of $(5x - 2)$ and $(2x + 7)$?

A $10x^2 + 31x + 14$

B $10x^2 + 39x - 14$

C $7x^2 + 31x - 14$

D $10x^2 + 31x - 14$

Yes! Well Done.

+0.50

Multiply each term in one polynomial by each term in the other polynomial:

$$\begin{aligned}\text{So } (5x - 2)(2x + 7) \\ &= 5x(2x + 7) - 2(2x + 7) \\ &= 10x^2 + 35x - 4x - 14 \\ &= 10x^2 + 31x - 14\end{aligned}$$

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Question 3

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Polynomials (Algebra 1, General)

[Help](#)Multiply out $(x - 4)(3x - y + 3)$

A $3x^2 + 4y - 12$

B $3x^2 - 6xy - 12$

C $3x^2 - xy - 9x + 4y - 12$

D $3x^2 - xy - 9x - 4y - 12$

Good try, but D is not right ... C is the right answer, read why:

+0.20

Multiply each term in one polynomial by each term in the other polynomial:

$$\begin{aligned}\text{So } (x - 4)(3x - y + 3) \\ &= x(3x - y + 3) - 4(3x - y + 3) \\ &= 3x^2 - xy + 3x - 12x + 4y - 12 \\ &= 3x^2 - xy - 9x + 4y - 12\end{aligned}$$

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Question 4

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Polynomials (Algebra 1, General)

[Help](#)What is the product of $(3x - 4y)$ and $(-2x + 5y - 6)$?

A $-6x^2 + 7xy - 18x - 20y^2 + 24y$

B $-6x^2 + 23xy - 18x - 20y^2 + 24y$

C $6x^2 + 23xy - 18x - 20y^2 + 24y$

D $-6x^2 + 23xy - 18x + 20y^2 - 24y$

Well Done!

+0.50

Multiply each term in one polynomial by each term in the other polynomial:

$$\begin{aligned}\text{So } (3x - 4y)(-2x + 5y - 6) \\ &= 3x(-2x + 5y - 6) - 4y(-2x + 5y - 6) \\ &= -6x^2 + 15xy - 18x + 8xy - 20y^2 + 24y \\ &= -6x^2 + 23xy - 18x - 20y^2 + 24y\end{aligned}$$

(57% answered this correctly)

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Question 5

[Next Question](#)

Polynomials (Algebra 1, General)

[Help](#)

What is the product of $(2y - 1)$, $(2y + 1)$ and $(4y^2 + 1)$?

A $16y^4 + 8y^2 - 1$

B $16y^4 - 8y^2 - 1$

C $16y^4 - 1$

D $8y^3 - 1$

Good try, but D is not right ... C is the right answer, read why:

+0.20

Step1 Find the product of the first two terms:

$$\begin{aligned}(2y - 1)(2y + 1) \\&= 2y(2y + 1) - 1(2y + 1) \\&= 4y^2 + 2y - 2y - 1 \\&= 4y^2 - 1\end{aligned}$$

Step2 Find the product of this with the third term:

$$\begin{aligned}(4y^2 - 1)(4y^2 + 1) \\&= 4y^2(4y^2 + 1) - 1(4y^2 + 1) \\&= 16y^4 + 4y^2 - 4y^2 - 1 \\&= 16y^4 - 1\end{aligned}$$

What is the product of $(3x - 2)$, $(2x + 3)$ and $(2x - 3)$?

A $12x^3 - 8x^2 - 27x + 18$

B $12x^3 - 8x^2 + 27x - 18$

C $12x^3 + 28x^2 + 3x - 18$

D $12x^3 - 28x^2 + 3x + 18$

Yes! That is Right!

+0.50

The product $(3x - 2)(2x + 3)(2x - 3)$ can be bracketed in two different ways:

$[(3x - 2)(2x + 3)](2x - 3)$ or
 $(3x - 2)[(2x + 3)(2x - 3)]$

The answers will be equal because multiplication is associative, but the second is easier since we can use the difference of two squares: $(2x + 3)(2x - 3) = (2x)^2 - 3^2 = 4x^2 - 9$

So:

$$\begin{aligned}(3x - 2)(2x + 3)(2x - 3) &= (3x - 2)[(2x + 3)(2x - 3)] \\ &= (3x - 2)(4x^2 - 9) \\ &= 12x^3 - 27x - 8x^2 + 18 \\ &= 12x^3 - 8x^2 - 27x + 18\end{aligned}$$

What is the product of $(2y^2 - 9)$, $(y + 3)$ and $(y - 3)$?

A $2y^4 + 12y^3 - 27y^2 + 81$

B $2y^4 - 27y^2 + 81$

C $2y^4 - 9y^2 + 81$

D $2y^4 + 9y^2 + 81$

Sorry, C is not right, the correct answer is B. Please see our solution:

+0.20

The product $(2y^2 - 9)(y + 3)(y - 3)$ can be bracketed in two different ways:

$(2y^2 - 9)[(y + 3)(y - 3)]$ or
 $[(2y^2 - 9)(y + 3)](y - 3)$

The answers will be equal because multiplication is associative, but the first is easier as we can use the difference of two squares:

$$(y + 3)(y - 3) = y^2 - 3^2 = y^2 - 9$$

So:

$$\begin{aligned}(2y^2 - 9)(y + 3)(y - 3) &= (2y^2 - 9)[(y + 3)(y - 3)] \\ &= (2y^2 - 9)(y^2 - 9) \\ &= 2y^4 - 18y^2 - 9y^2 + 81 \\ &= 2y^4 - 27y^2 + 81\end{aligned}$$

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Question 8

[Next Question](#)

Polynomials (Algebra 1, Hard)

[Help](#)

What is the product of $(3a - 4b + 1)$ and $(2a + 5b - 3)$?

A $6a^2 + 20b^2 + 7ab - 7a + 17b - 3$

B $6a^2 - 20b^2 + 7ab - 7a + 17b - 3$

C $6a^2 - 20b^2 + 23ab - 7a + 17b - 3$

D $6a^2 - 20b^2 + 7ab - 7a - 7b - 3$

Good try, but D is not right ... B is the right answer, read why:

+0.20

Multiply each term in one polynomial by each term in the other polynomial:

$$\begin{aligned}(3a - 4b + 1)(2a + 5b - 3) \\&= 3a(2a + 5b - 3) - 4b(2a + 5b - 3) + 1(2a + 5b - 3) \\&= 6a^2 + 15ab - 9a - 8ab - 20b^2 + 12b + 2a + 5b - 3 \\&= 6a^2 - 20b^2 + 7ab - 7a + 17b - 3\end{aligned}$$

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Question 9

Polynomials (Algebra 2, Hard)

[Help](#)

Simplify $(2x + y)(x - 2y) - (x + 2y)(2x - y)$ as much as possible.

A $6xy$ B $4x^2 - 6xy - 4y^2$ C $4x^2 - 4y^2$ D $-6xy$

Good try, but B is not right ... D is the right answer, read why:

+0.20

First simplify each of the products:

1. $(2x + y)(x - 2y)$

Multiply each term in one polynomial by each term in the other polynomial:

So $(2x + y)(x - 2y)$

$$= 2x(x - 2y) + y(x - 2y)$$

$$= 2x^2 - 4xy + xy - 2y^2$$

$$= 2x^2 - 3xy - 2y^2$$

2. $(x + 2y)(2x - y)$

Multiply each term in one polynomial by each term in the other polynomial:

So $(x + 2y)(2x - y)$

$$= x(2x - y) + 2y(2x - y)$$

$$= 2x^2 - xy + 4xy - 2y^2$$

$$= 2x^2 + 3xy - 2y^2$$

Now subtract :

$$(2x + y)(x - 2y) - (x + 2y)(2x - y)$$

$$= 2x^2 - 3xy - 2y^2 - (2x^2 + 3xy - 2y^2)$$

$$= 2x^2 - 3xy - 2y^2 - 2x^2 - 3xy + 2y^2 \text{ (remembering to change signs)}$$

$$= 2x^2 - 2x^2 - 3xy - 3xy - 2y^2 + 2y^2$$

$$= -6xy$$